La Scuola d'Italia / Dr. Huson / IB Math 11
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Nickname:

2.9 Classwork: Every Rate of Change Has a Name

In every field, the slope (*pendenza, coefficiente angolare*) of a line gets its own name, and the intercept (*intercetta, ordinata all'origine*) means something too. These models show how things work, but real life is more complicated. All figures are simplified.

1. Sales tax in New York City. The sales tax rate (*aliquota*) is 8.875%. For a price x dollars before tax, the amount paid is $P = 1.08875x$. The graph is a line through the origin: a direct proportion (*proporzionalità diretta*). Its slope, 1.08875, is the constant of proportionality (*costante di proporzionalità*).

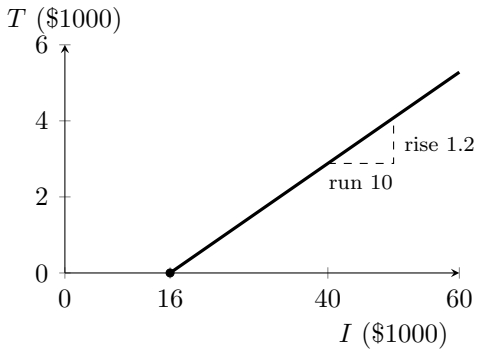
(a) A receipt shows a total of \$174.20. Find the price before tax.

2. Income tax, one bracket.

Simplified US model: no tax on the first \$16 000 of income (the standard deduction (*deduzione standard*)), then 12% on the rest.

$$T = 0.12(I - 16\,000), \quad I \geq 16\,000.$$

The slope, 0.12, is the **marginal tax rate** (*aliquota marginale*): the tax on one more dollar. The deduction is the I-intercept, not the T-intercept.



(a) Find the tax on an income of \$40 000.

(b) The **average tax rate** (*aliquota media*) is $T \div I$. Find it for $I = 40\,000$ and for $I = 100\,000$. Will it ever reach 12%? Is the graph of the average rate a straight line?

(c) A *tax credit* (*credito d'imposta, detrazione*) of \$500 subtracts 500 from T. A larger deduction of \$20 000 replaces 16 000. Describe how each one moves the graph.

3. A budget line (*vincolo di bilancio*) on Mulberry Street. You have \$24 to spend on espressos at \$3 and cannoli at \$4. With x espressos and y cannoli, $3x + 4y = 24$. The slope is $-\frac{3}{4}$, a **negative** rate of change: each extra espresso costs $\frac{3}{4}$ of a cannolo. This is the relative price (*prezzo relativo*), or opportunity cost (*costo opportunità*).

(a) Find both intercepts and say what each one means.

(b) The price of a cannolo rises to \$6. Write the new budget line and find its slope.

4. Break-even at an Arthur Avenue pizzeria. Fixed costs (*costi fissi*) are \$9000 a month. Each pizza costs \$6 to make and sells for \$18. For q pizzas a month, cost is $C = 9000 + 6q$ and revenue (*ricavo*) is $R = 18q$. The slope of C is the **marginal cost** (*costo marginale*). The slope of the profit $R - C$ is the **contribution margin** (*margine di contribuzione*).

(a) Find the break-even point (*punto di pareggio*), where $C = R$.

(b) Find the profit when the pizzeria sells 1000 pizzas in a month.

5. A tariff (*dazio*) on imported Parmigiano. In the US market, demand is $p = 40 - 2q$ and supply is $p = 16 + q$, where p is dollars per kg and q is thousand kg per week. The demand slope, -2, is the **price sensitivity** (*sensibilità al prezzo*). (Economists also use *elasticity* (*elasticità*): the same idea in percent terms, % change in quantity per % change in price.)

(a) Use elimination to find the equilibrium quantity and price.

(b) A tariff of \$3 per kg shifts the supply line up by 3. Find the new equilibrium.

(c) How much of the \$3 tariff do buyers pay? This share is the **pass-through** (*traslazione dell'imposta*).

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6. Two ways to pay tax. Some countries, Italy among them (*regime forfettario*), let a freelancer choose a flat-rate tax on revenue. Simplified, for revenue R dollars:

Flat rate: $T = 0.15R$ (slope: the **effective rate** (*aliquota effettiva*))
Standard: $T = 0.25(R - 20\,000)$ (slope: the marginal rate)

(a) Use elimination to find the revenue at which the two options cost the same. Which option is cheaper for $R = 80\,000$?

7. Stocks and the market. A simple model links a stock's monthly return (*rendimento*) r to the market's return r_m (both in %): $r = \alpha + \beta\, r_m$. The slope β is the **beta** (*coefficiente beta*): how many % the stock moves per 1% move of the market. The intercept α (*alfa*) is the return when the market is flat.

A bank stock: $r = 0.5 + 1.5\, r_m$
Con Edison (a utility): $r = 0.2 + 0.5\, r_m$
An inverse fund (*ETF inverso*): $r = -r_m$ ($\beta = -1$: a **negative** slope)

(a) The market falls 4% in a month. Find the return on each of the three.

(b) You split \$10 000 between the bank (\$$x$) and Con Edison (\$$y$) so that the portfolio has a beta of 1.2: $1.5x + 0.5y = 1.2 \times 10\,000$. Find x and y.

8. Bonds (*obbligazioni*): price falls when yield rises. When interest rates (yields, (*rendimento a scadenza*)) rise, the price of a bond falls: a **negative** slope. The **duration** (*durata finanziaria*) is that slope in percent terms: a duration of 7 means the price falls about 7% when the yield rises by 1 percentage point. For a US Treasury bond priced at \$100 when the yield is 4%: $P \approx 100 - 7(y - 4)$, with y in %.

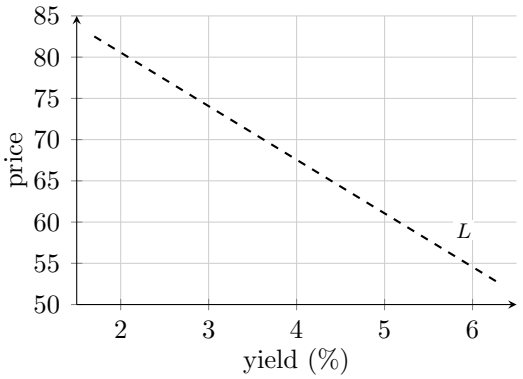
(a) Estimate the price if the yield rises to 4.5%.

(b) You split \$100 000 between a short bond (duration 2) and a long bond (duration 7) to get a duration of 5: $2x + 7y = 5 \times 100\,000$. Find the amount in each bond.

When the rate of change changes: curves. In Example 2(b) the average tax rate was not a straight line. When the slope itself changes, we need curves. The way the slope changes also has a name.

9. Convexity (*convessità*). A 10-year zero-coupon bond pays \$100 in 10 years. Its price is the present value $P = \dfrac{100}{(1+y/100)^{10}}$, where y is the yield in %. The straight line is the duration estimate at 4%: $L = 67.56 - 6.50(y-4)$.

Yield (%)	Price P	Line L
2		80.55
3		74.05
4	67.56	67.56
5		61.06
6		54.56



(a) Use your GDC to complete the table. Plot the prices and join them with a smooth curve.

(b) Is the curve above or below the line? Does the curve get steeper or less steep as the yield rises?

10. Economies of scale (*economie di scala*): the learning curve (*curva di apprendimento*). A factory's first battery pack costs \$10 000. Each time the number of packs made doubles, the cost per pack falls by 20% (the **learning rate**).

(a) Find the cost of pack number 2, 4, 8, 16 and 32. Which pack is the first to cost less than \$4500?

11. The S-curve (*curva logistica*). Percent of US households owning a new product, t years after launch (simplified):

t	0	1	2	3	4	5	6	7	8
%	2	5	12	27	50	73	88	95	98
1st difference									

(a) Complete the first differences. When is the product spreading fastest? This is the **inflection point** (*punto di flesso*), where the curve changes from bending up to bending down.